BIS SCIENCE PROJECTS 2026 — CLASS 7 (EV)

# EARTHQUAKE & TREMOR ALARM SYSTEM

An IoT-Enabled Hardware-Interrupt Early Warning System

System Architecture & Hardware Specification Blueprint

## PROJECT TEAM ROSTER & ROLE ALLOCATIONS

**Abdullah Karim**
PROJECT LEAD • SYSTEM ARCHITECT • CODER • BUILDER • PLANNER

**Mahir**
ASST. BUILDER & MAIN CONTRIBUTOR

**Ahmad**
MAIN CONTRIBUTOR & PLANNER

**Behan**
MAIN CONTRIBUTOR & PLANNER

**Zarif**
CONTRIBUTOR

## ACADEMIC & PROJECT OVERVIEW

| | |
|---|---|
| **Academic Event** | BIS Science Projects 2026 / Class 7 English Version (EV) |
| **Project Author & Lead** | **Abdullah Karim** (System Architect & Lead Engineer) |
| **Project Objective** | Design and build an earthquake early warning system that triggers visual/acoustic alarms locally and sends remote alerts over Wi-Fi within 15ms. |

## COMPLETE BILL OF MATERIALS (BOM) & COMPONENT SOURCING

| # | COMPONENT | TECHNICAL SPECIFICATION | QTY | USD COST | BDT COST |
|---|---|---|---|---|---|
| 1 | **ESP32 Board** | ESP-WROOM-32 (30-Pin NodeMCU, Dual Core, CP2102) | 1 | $0.00 (Owned) | ৳0.00 (Owned) |
| 2 | **SW-420 Sensor** | Vibration Sensor Module with LM393 Comparator | 1 | $1.20 | ৳145.00 |
| 3 | **Active Buzzer** | 5V/3.3V Sound Transducer, Sound Level >85dB | 1 | $0.40 | ৳50.00 |
| 4 | **BC547 NPN** | TO-92 NPN Transistor (45V 100mA Switch) | 1 | $0.10 | ৳12.00 |
| 5 | **1 kΩ Resistor** | 1/4W Metal Film, ±5% (Transistor Base Protection) | 1 | $0.02 | ৳3.00 |
| 6 | **330 Ω Resistor** | 1/4W Metal Film, ±5% (LED Current Limiter) | 1 | $0.02 | ৳3.00 |
| 7 | **Red LED (5mm)** | High-brightness Diffused Red LED (Vf = 2.0V) | 1 | $0.08 | ৳10.00 |
| 8 | **Breadboard** | MB-102 830-Point Solderless Breadboard | 1 | $1.50 | ৳180.00 |
| 9 | **Jumper Wires** | 20cm Male-to-Male Jumpers Set | 10 | $0.38 | ৳45.00 |
| | | **TOTAL OUT-OF-POCKET COST:** | **17** | **$3.70 USD** | **৳448.00 BDT** |

ABDULLAH KARIM

# EARTHQUAKE & TREMOR ALARM SYSTEM

**WIRING SPECIFICATION & ASSEMBLY WALKTHROUGH**

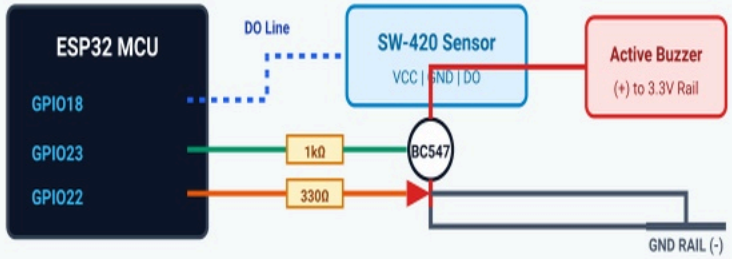
Doc ID: ETA-2026-SPEC-V5
Revision: 5.0.0 (Final)

| TRIGGER RESPONSE | OPERATING VOLTAGE | ACOUSTIC OUTPUT | ADJUSTED TOTAL COST |
|---|---|---|---|
| < 15 ms | 3.3V DC | 85 dBA | $3.70 USD |
| Hardware Interrupt | 5V USB Input | @ 10cm Distance | ~448 BDT (ESP32 Owned) |

## HARDWARE MASTER WIRING SPECIFICATION MATRIX

| PIN ID | WIRE COLOR | DESTINATION COMPONENT | LOGIC / RAIL | BOARD COORDINATES |
|---|---|---|---|---|
| **GPIO 18** | Blue Cable | SW-420 Signal Pin (DO) | 3.3V DIGITAL IN | ESP Pin 18 → Row 23 Pin DO |
| **GPIO 23** | Green Cable | 1kΩ Resistor → BC547 Base | 3.3V DIGITAL OUT | ESP Pin 23 → Row 27 (Base) |
| **GPIO 22** | Orange Cable | 330Ω Resistor → LED Anode | 3.3V DIGITAL OUT | ESP Pin 22 → Row 30 (Anode) |
| **3V3** | Red Cable | Breadboard Power Rail (+) | +3.3V DC RAIL | ESP 3V3 Pin → Bus Rail (+) |
| **GND** | Black Cable | Breadboard Ground Rail (-) | 0V GROUND RAIL | ESP GND Pin → Bus Rail (-) |

## SYSTEM SCHEMATIC ARCHITECTURE (VECTOR DIAGRAM)




## COMPREHENSIVE 8-STEP PHYSICAL CONSTRUCTION GUIDE

### 1 Mount ESP32 Module
- Place ESP32 across central divide ditch (Rows 1–15).
- Ensure Pin 3V3 sits on Row 1 (Left) and Vin on Row 1 (Right).

### 2 Power Rails setup
- Connect jumper wire from ESP32 **3V3** to Red (+) Rail.
- Connect jumper wire from ESP32 **GND** to Black (-) Rail.

### 3 Position Sensor
- Mount SW-420 sensor vertically on breadboard Rows 20–23.
- Jumper SW-420 **VCC** to 3.3V rail and **GND** to ground rail.

### 4 Sensor Calibration
- Connect SW-420 **DO** pin to ESP32 **GPIO 18**.
- Turn blue potentiometer counter-clockwise to adjust trigger limit.

### 5 Install Transistor
- Insert BC547 on Rows 26–28; wire **1kΩ resistor** from **GPIO 23** to Middle Pin.
- Connect Right Pin (Emitter) to ground rail (-).

### 6 Connect Buzzer
- Connect buzzer (+) anode to 3.3V power rail (+).
- Connect (-) cathode to BC547 Left Pin (Collector).

### 7 Status LED Assembly
- Bridge **330Ω resistor** from **GPIO 22** to LED anode Row 30.
- Connect LED cathode directly to ground rail (-).

### 8 Power & Verification
- Inspect wiring. Plug USB to ESP32; confirm power LED glows.
- Tap sensor to test immediate local buzzer & LED alert.

**Power Supply & Protection Note:** The system is powered by a standard 5V USB source (phone charger or power bank) plugged into the ESP32. Transistor switching isolates piezoceramic voltage spikes from ruining MCU GPIO digital pins.

ABDULLAH KARIM

# FIRMWARE ARCHITECTURE & REMOTE ALERT DISPATCH

## HARDWARE INTERRUPT & NETWORK GATEWAYS

Hardware ISR & Software Backends

## CORE HARDWARE INTERRUPT LOGIC (IRAM_ATTR ISR)

```
// Optimized Hardware ISR Logic Implementation — Abdullah Karim #include <Arduino.h> #define SENSOR_PIN 18 #define BUZZER_PIN 23 #define LED_PIN 22 volatile bool tremorDetected = false; unsigned long lastTriggerTime = 0; void IRAM_ATTR sensorISR() { unsigned long currentTime = millis(); if (currentTime - lastTriggerTime > 500) { // 500ms debounce tremorDetected = true; lastTriggerTime = currentTime; } } void setup() { pinMode(SENSOR_PIN, INPUT_PULLUP); pinMode(BUZZER_PIN, OUTPUT); pinMode(LED_PIN, OUTPUT); attachInterrupt(digitalPinToInterrupt(SENSOR_PIN), sensorISR, FALLING); }
```

## COMPREHENSIVE REMOTE NOTIFICATION GATEWAYS (5 BACKENDS)

### 1. Telegram HTTPS Bot

SSL REST

```
#include <WiFiClientSecure.h> void sendTelegramAlert(String token, String chat) { WiFiClientSecure client; client.setInsecure(); if (client.connect("api.telegram.org", 443)) { client.print("GET /bot" + token + "/sendMessage?chat_id=" + chat + "&text=TREMOR_ALERT HTTP/1.1\r\nHost: api.telegram.org\r\n\r\n"); } }
```

### 2. WhatsApp CallMeBot

REST API

```
#include <HTTPClient.h> void sendWhatsAppAlert(String phone, String key) { HTTPClient http; http.begin("https://api.callmebot.com/whatsapp.php?phone=" + phone + "&text=TREMOR_DETECTED&apikey=" + key); http.GET(); http.end(); }
```

### 3. Twilio Cellular SMS

BASIC AUTH

```
HTTPClient http; http.begin("https://api.twilio.com/2010-04-01/Accounts/" + sid + "/Messages.json"); http.setAuthorization(sid.c_str(), token.c_str()); http.POST("To=" + targetNum + "&From=" + twilioNum + "&Body=EARTHQUAKE_ALERT");
```

### 4. Async WebSockets

< 15MS LOCAL

```
#include <ESPAsyncWebServer.h> AsyncWebSocket ws("/ws"); void triggerLocalBroadcast() { if (ws.count() > 0) ws.textAll("{\"status\":\"CRITICAL_TREMOR\"}"); }
```

### 5. Encrypted SMTP Email Gateway

TLS PORT 465

```
#include <ESP_Mail_Client.h> SMTPSession smtp; Session_Config config; config.server.host_name = "smtp.gmail.com"; config.server.port = 465; config.login.email = SENDER_EMAIL; config.login.password = APP_PASSWORD; // Connects via SSL and dispatches detailed event telemetry report
```

## FLASHING & DEPLOYMENT MANUAL

1. Open Arduino IDE (v2.0+) and add ESP32 URL in Preferences.
2. Install ESP32 Board Manager package and select **ESP32 Dev Module**.
3. Install `ArduinoJson`, `ESPAsyncWebServer`, and `ESP-Mail-Client` libraries.
4. Set Upload Speed to 921600 and CPU Frequency to 240MHz. Flash firmware.